

Data Analytics Insights Report

Part 1:

Task 1: SQL-Based Traffic Analysis

Analyze annual traffic trends

- Identify years with increases or decreases. Years with increase in traffic volume – 2013,2016,2017
Years with decrease in traffic volume – 2014,2015,2018

Task 2: Descriptive Statistics and Correlation

This descriptive statistics summary reveals critical characteristics about the **traffic volume distribution, variability, and data structural biases** .

- a) The Average Vs. The Middle: The Mean traffic volume is 3259.82, while the Median is higher at 3380.
- b) The Anomalous Mode: The Mode is 353, which is vastly lower than both the mean and median. In traffic volume analysis, this highly repetitive value usually indicates a persistent data artifact—such as a specific sensor clipping threshold, a recurring baseline nighttime volume, or a placeholder for missing/corrupted telemetry.
- c) Massive Volatility: The Standard Deviation is 1986.86 against a mean of 3259.82. This results in a remarkably high Coefficient of Variation (~61%), showing that traffic is highly volatile and fluctuates drastically between empty roads and severe congestion.
- d) The Maximum Spread: The Range is 7280, stretching from a Minimum of 0 (complete absence of traffic or sensor downtime) to a Maximum of 7280 vehicles.
- e) Platykurtic Distribution: The Kurtosis is -1.31. A negative kurtosis value signifies a "flat-topped" distribution with shorter, thinner tails. This indicates that traffic volume doesn't have extreme isolated spikes; rather, it frequently fluctuates broadly across its entire range
- f) Tight Confidence Intervals: Due to the massive sample size, the Standard Error is very low (9.05) , yielding a narrow 95% Confidence Level margin of error ..

Task 3: Probability and Congestion Analysis

The probability data yields three main conclusions:

- **Low, Cyclical Baseline Risk:** The probability of congestion during clear weather is generally low and highly predictable. It clusters almost exclusively around fixed, peak commuter hours (morning and evening rush hours), making it easy for predictive models to manage.
- **Maximum Buffer Capacity:** Clear weather columns represent periods where roads operate at their true engineered capacity. Because driver visibility, braking distance, and

reaction times are optimal, the network possesses the maximum physical buffer to absorb micro-surges in volume without collapsing into gridlock.

- **Linear vs. Non-Linear Friction:** While an increase in traffic volume during clear weather leads to a predictable, linear rise in travel times, the data columns for adverse weather (like rain or snow) show a non-linear spike in congestion probability. Poor weather drops roadway capacity by 10% to 30%, meaning a volume level that flows perfectly under a "Clear" column will cause total gridlock under a "Rain" column.

Task 4: Power BI Traffic Intelligence Dashboard

4.2 Dashboard analysis

D. Temperature and traffic

Create a scatter plot of:

- Temperature vs. traffic volume.

Discuss:

- Whether a visible relationship exists.

Observation: The plot shows a thick, **dense vertical block** of blue data points concentrated on the right side of the chart.

Analysis: There is **no visible linear relationship (no correlation)** between temperature and traffic volume. The dots do not form an upward trend (positive correlation) or a downward trend (negative correlation). This indicates that changes in temperature do not directly cause traffic volume to rise or fall predictably. Traffic remains spread across all levels regardless of the temperature.

- The temperature range associated with higher traffic.

Observation: The bulk of the high traffic volume (the tallest part of the cluster along the Y-axis) is heavily concentrated on the X-axis between roughly **260 K and 300 K**.

Analysis: Converting this from Kelvin to Celsius, the peak traffic volume occurs within a temperature range of approximately **-13°C to 27°C**. This represents the standard atmospheric operating range for this region where daily commuting takes place normally.

- Any notable outliers.

Observation: There is a distinct, isolated vertical line of scattered dots on the extreme left side of the chart, sitting right at **0 K** on the X-axis.

Analysis: These are **highly significant outliers**. A temperature of 0 Kelvin translates to **-273.15°C** (Absolute Zero), which is physically impossible in a real-world traffic environment. These data points represent corrupted or missing data (system errors where unrecorded temperatures defaulted to zero) and should be cleaned out before final reporting